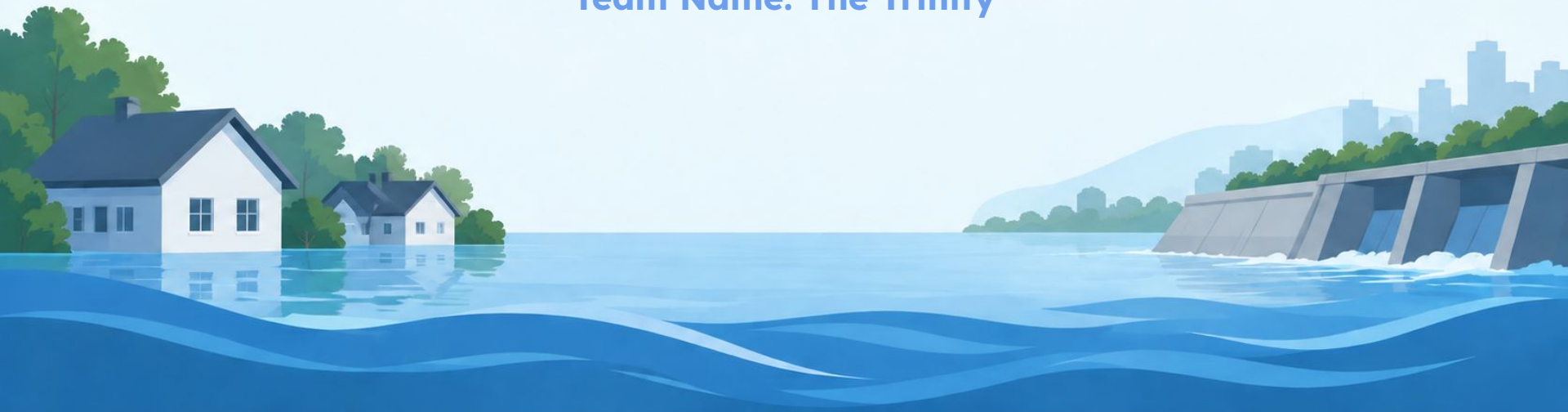


PROJECT-FloodRecon

Real-life Hazard Detection from Aerial Drone Footage

Team Name: The Trinity



The Critical Challenge in Flood Response

- Rescue teams receive massive amounts of drone footage but lack the time to analyse it manually.
- Human analysis is slow, subjective, and misses critical details in chaotic environments.
- **The Risk:** First responders waste precious minutes identifying where to go first leading to preventable delays.

Our Solution - FloodRecon AI

How FloodRecon AI Works

- We deploy a drone to capture footage of the flooded region.
- Our AI processes the video frame-by-frame in real-time.
- **The Output:** The system returns the media with hazards clearly outlined in color-coded boxes, turning raw video into actionable intelligence instantly.

The Core Technology we use to achieve this - [YOLOv8](#)

Why YOLOv8 ?

- **Object Detection:** We detect specific hazards like Collapsed Electric Poles, Waterlogging, and Cracked Walls.
- **Speed:** YOLO processes frames in milliseconds - perfect for live drone feeds.
- **Accuracy:** Industry-standard reliability for distinguishing objects in cluttered, flood-affected landscapes.
- **Prototyping:** User-friendly for quick development for the Hackathon.

RED, YELLOW, GREEN - Prioritizing Action

- **RED (DANGER - Immediate Action):** Collapsed electric poles, submerged vehicles, or structural collapses. *Evacuate Immediately.*
- **YELLOW (Liability - Monitor Closely):** Cracks in walls/buildings, tilting structures. *Risk of collapse if water rises.*
- **GREEN (Relatively Safe - Accessible):** Minor waterlogging, flooded roads with low depth. *Safe for rescue boats and foot traffic.*

Training the AI - Our Data Strategy

- **Public Datasets:** We utilize pre-recorded UAV footage of flood-prone areas (inspired by FloodAI & Hermes AI).
- **Data Augmentation:** We use python libraries to artificially multiply our dataset by rotating, flipping, and adjusting brightness - making the AI robust to different weather and lighting.
- **Labelling:** We strictly define our hazard classes: `collapsed_pole`, `building_crack`, `fooded_road` to ensure the AI knows exactly what to look for.

Implementation Blueprint

- **Framework:** Python with **PyTorch** for deep learning.
- **Dataset Management: Roboflow** - to handle labelling, augmentation, and slitting our data in Training, Validation, and Test sets.
- **The Model: Ultralytics YOLOv8** - pre-trained on **COCO** and fine-tuned specifically on our disaster classes.
- **Post-Processing:** A custom script that assigns the Red/Yellow/Green classification based on the detected object and confidence score.

From Drone to Decision (The Loop)

- **Input (Drone Feed):** The drone streams live video or sends a pre-recorded .mp4 file to our system.
- **Frame Extraction (OpenCV):** We use **OpenCV** to read the video file/stream frame-by-frame for the AI to analyze.
- **Object Detection (YOLOv8):** Each extracted frame is passed into our YOLOv8 model, which detects hazards like collapsed poles, cracks, and waterlogging.
- **Color Classification:** Based on the detected object, our script assigns the appropriate color:
- **Output (OpenCV + Media):** OpenCV draws the colored bounding boxes directly onto the original frames. The system then outputs a new annotated video or a collage of key frames, ready for rescue teams to view instantly.

Key Resources & Inspiration

- **FloodAI:** For advanced UAV flood image segmentation and reporting.
- **Hermes AI:** For UAV-based damage mapping and infrastructure assessment.
- **Flood Disaster Detection (GitHub):** For proof-of-concept YOLO flood scene recognition.
- **Takeaway:** Leveraging these open-source frameworks allows us to build a working prototype for the Hackathon in a timely fashion.

Impact & Future Vision

- **Immediate Impact:** Cuts hazard assessment time from hours to seconds. Command centers can prioritize rescue routes instantly.

Future Scope :-

- **Object Tracking:** If a pole disappears between frames, the AI triggers an instant “Collapse Alert”.
- **Flood Mapping:** Transitioning from bounding boxes to pixel-perfect segmentation to calculate exact flood volumes.

THANK YOU